**Cambridge International Examinations**
Cambridge International General Certificate of Secondary Education

**CHEMISTRY** **0620/23**

Paper 2 Multiple Choice (Extended) **May/June 2016**

**45 Minutes**

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
A copy of the Periodic Table is printed on page 20.
Electronic calculators may be used.

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **17** printed pages and **3** blank pages.

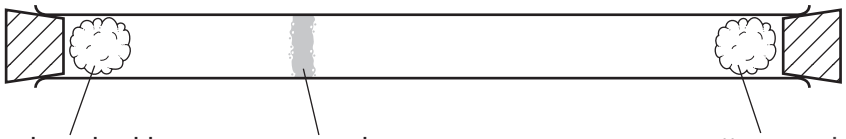
IB16 06_0620_23/3RP
© UCLES 2016

CAMBRIDGE International Examinations

[Turn over

www.root19.com +91 - 9969 353 391

**1** The diagram shows an experiment to demonstrate diffusion.




Which statement explains why the ring of ammonium chloride appears as shown?

**A** Ammonia solution only produces a gas which moves until it meets the hydrochloric acid.

**B** Both solutions produce a gas, but ammonia moves quicker than hydrogen chloride because it is lighter.

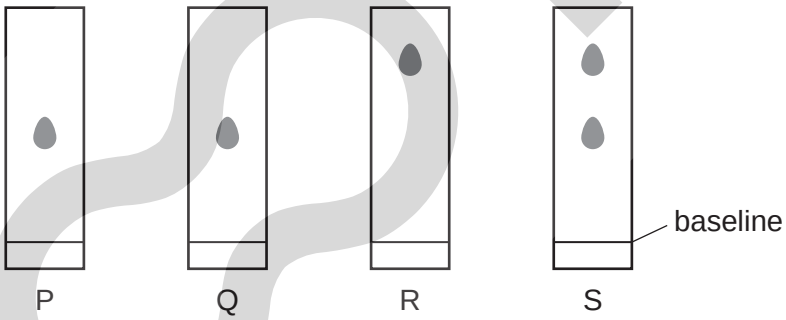
**C** Hydrochloric acid produces hydrogen chloride which stays at one end of the tube until the ammonia reaches it.

**D** The two solutions run along the tube until they meet.

**2** Chromatography experiments are carried out on four substances, P, Q, R and S.

The same solvent is used in each experiment.

The resulting chromatograms are shown below.




Which statement is **not** correct?

**A** P and Q are pure substances.

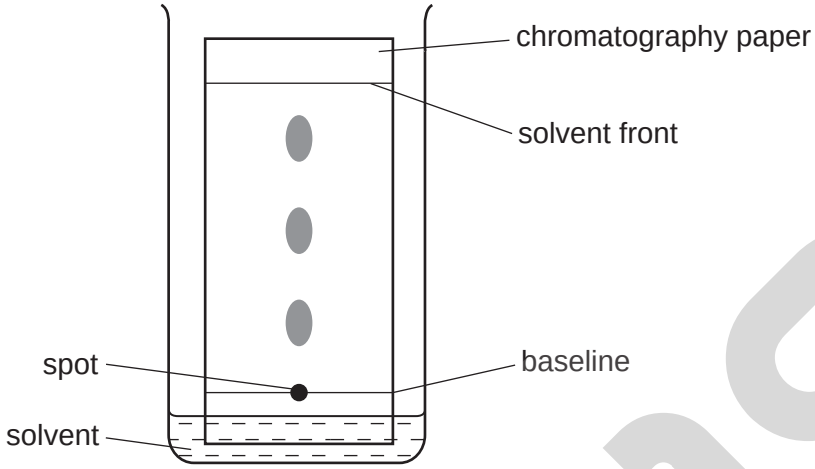
**B** P and R are different substances.

**C** R and S are pure substances.

**D** S is a mixture of substances.


www.root19.com +91 - 9969 353 391

**3** The diagram shows the apparatus used to separate the different components of a mixture by chromatography.




Which statement about this experiment is correct?

**A** A locating agent is used to find the position of the solvent front.

**B** The components to be separated must be soluble in the solvent.

**C** The baseline on which the spot of the mixture is placed is drawn in ink.

**D** The $R_f$ value is calculated by $\frac{\text{the distance travelled by the solvent front}}{\text{the distance travelled by the component}}$

**4** Which statements about isotopes of the same element are correct?

1 They are atoms which have the same chemical properties because they have the same number of electrons in their outer shell.

2 They are atoms which have the same number of electrons and neutrons but different numbers of protons.

3 They are atoms which have the same number of electrons and protons but different numbers of neutrons.

**A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

root 19


www.root19.com +91 - 9969 353 591

**5** The table shows the electronic structure of four atoms.

| atom | electronic structure |
|---|---|
| W | 2,8,1 |
| X | 2,8,4 |
| Y | 2,8,7 |
| Z | 2,8,8 |

Which two atoms combine to form a covalent compound?

**A** W and X **B** W and Y **C** X and Y **D** X and Z

**6** Which statement describes the attractive forces between molecules (intermolecular forces)?

**A** They are strong covalent bonds which hold molecules together.

**B** They are strong ionic bonds which hold molecules together.

**C** They are weak forces formed between covalently-bonded molecules.

**D** They are weak forces which hold ions together in a lattice.

**7** Which substance exists as a lattice of positive ions in a 'sea of electrons'?

**A** liquid potassium chloride

**B** solid graphite

**C** solid magnesium

**D** solid silicon(IV) oxide

**8** Analysis of a compound formed between magnesium and nitrogen showed it contained 14.4 g of magnesium and 5.6 g of nitrogen.

What is the empirical formula of the compound?

**A** $Mg_2N_3$ **B** $Mg_3N_2$ **C** $Mg_4N_6$ **D** $Mg_6N_4$

**9** An excess of zinc is added to 100 $cm^3$ of 1.0 mol / $dm^3$ hydrochloric acid.

The equation for the reaction is:

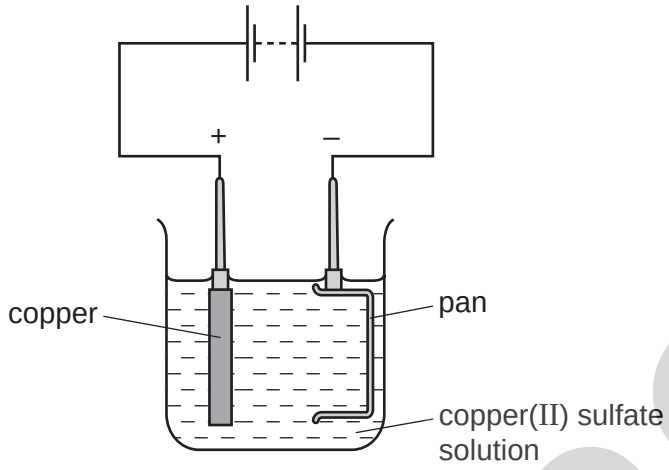
$$Zn + 2HCl \rightarrow ZnCl_2 + H_2$$

What is the maximum volume of hydrogen evolved at room temperature and pressure?

**A** 1.2 $dm^3$ **B** 2.0 $dm^3$ **C** 2.4 $dm^3$ **D** 24 $dm^3$


www.root19.com +91 - 9969 353 391

**10** The diagram shows a method used to copper-plate a pan




Which equation represents the reaction at the cathode?

**A** $Cu^{2+} + 2e^- \rightarrow Cu$

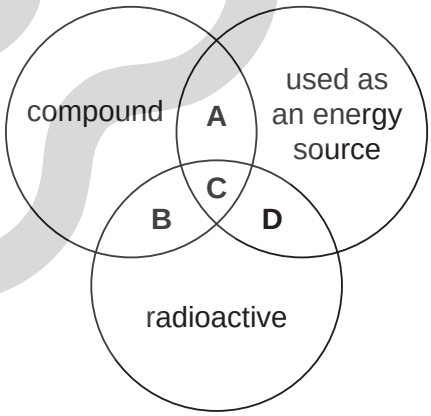
**B** $2H^+ + 2e^- \rightarrow H_2$

**C** $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

**D** $2O^{2-} \rightarrow O_2 + 4e^-$

**11** The diagram shows some properties that substances may have.

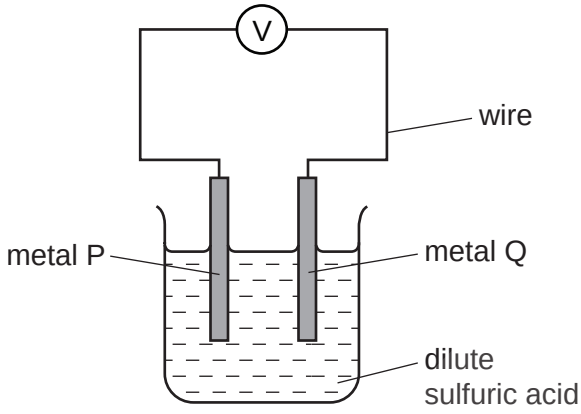
To which labelled part of the diagram does $^{235}U$ belong?





www.root19.com +91 - 9969 353 591

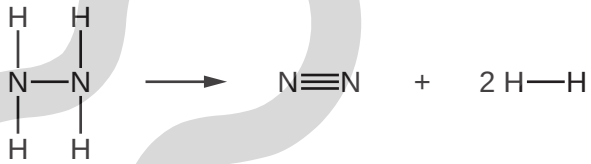
**12** The diagram shows a simple cell.




Which pair of metals produces the largest voltage?

| | metal P | metal Q |
|---|---|---|
| **A** | iron | copper |
| **B** | magnesium | copper |
| **C** | magnesium | zinc |
| **D** | zinc | copper |

**13** Hydrazine, $N_2H_4$, decomposes as shown.




The energy change for this reaction is –95 kJ / mol.

The table shows some bond energies involved.

| bond | bond energy in kJ / mol |
|---|---|
| N≡N | 945 |
| N–H | 391 |
| H–H | 436 |

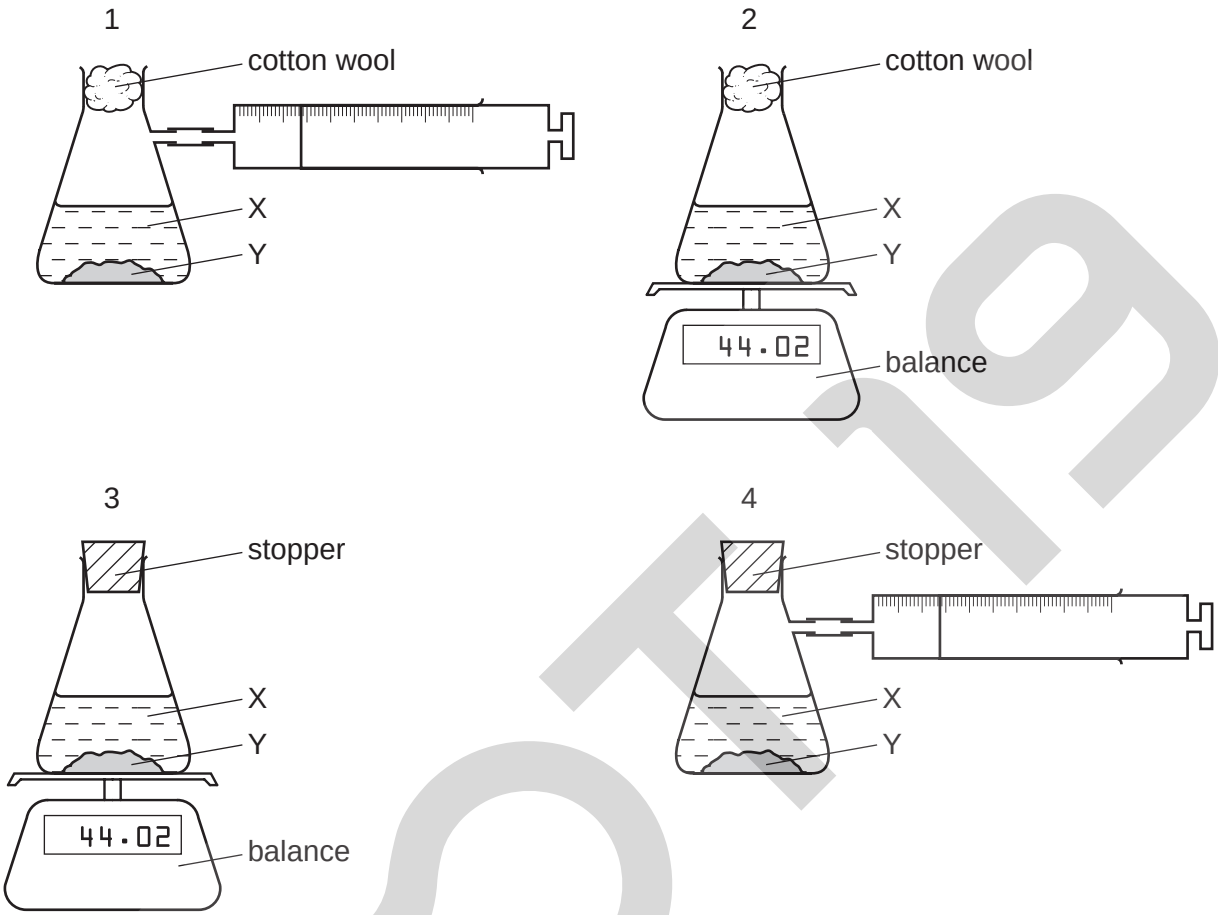
What is the bond energy of the N–N bond?

**A** 158 kJ / mol **B** 315 kJ / mol **C** 348 kJ / mol **D** 895 kJ / mol


www.root19.com +91 - 9969 353 391

**14** A liquid X reacts with solid Y to form a gas.

Which two diagrams show suitable methods for investigating the rate (speed) of the reaction?




**A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

**15** Which row explains why increasing temperature increases the rate of reaction?

| | particles collide more often | particles collide with more energy |
|---|---|---|
| **A** | ✓ | ✓ |
| **B** | ✓ | ✗ |
| **C** | ✗ | ✓ |
| **D** | ✗ | ✗ |


www.root19.com +91 - 9969 353 591

**16** Methanol is manufactured by reacting carbon monoxide and hydrogen together in the presence of an aluminium oxide catalyst.

The equation for the reaction is shown.

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

The reaction is a reversible reaction.

The forward reaction is exothermic.

Which change in conditions increases the yield of methanol?

**A** decreasing the concentration of the carbon monoxide

**B** increasing the pressure

**C** increasing the rate of the reaction

**D** increasing the temperature

**17** Which equation represents a reduction reaction?

**A** $Fe^{2+} + e^- \rightarrow Fe^{3+}$

**B** $Fe^{2+} \rightarrow Fe^{3+} + e^-$

**C** $Fe^{3+} + e^- \rightarrow Fe^{2+}$

**D** $Fe^{3+} \rightarrow Fe^{2+} + e^-$

**18** Which statements are properties of an acid?

1 reacts with ammonium sulfate to form ammonia

2 turns red litmus blue

| | 1 | 2 |
|---|---|---|
| **A** | ✓ | ✓ |
| **B** | ✓ | ✗ |
| **C** | ✗ | ✓ |
| **D** | ✗ | ✗ |


www.root19.com +91 - 9969 353 391

**19** Which row describes whether an amphoteric oxide reacts with acids and bases?

| | reacts with acids | reacts with bases |
|---|---|---|
| **A** | no | no |
| **B** | no | yes |
| **C** | yes | no |
| **D** | yes | yes |

**20** Barium sulfate is an insoluble salt.

It can be made by reacting copper(II) sulfate solution with barium nitrate solution.

$$CuSO_4(aq) + Ba(NO_3)_2(aq) \rightarrow Cu(NO_3)_2(aq) + BaSO_4(s)$$

What is the correct order of steps to obtain a pure, dry sample of barium sulfate from the reaction mixture?

| | step 1 | step 2 | step 3 |
|---|---|---|---|
| **A** | filter | evaporate the filtrate to dryness | leave the solid formed to cool |
| **B** | filter | evaporate the filtrate to the point of crystallisation | leave the filtrate to cool |
| **C** | filter | leave the residue in a warm place to dry | wash the residue with water |
| **D** | filter | wash the residue with water | leave the residue in a warm place to dry |

**21** Where in the Periodic Table is the metallic character of the elements greatest?

| | left or right side of a period | at the top or bottom of a group |
|---|---|---|
| **A** | left | bottom |
| **B** | left | top |
| **C** | right | bottom |
| **D** | right | top |


www.root19.com +91 - 9969 353 591

**22** Which statement about the elements in Group I is correct?

**A** Hydrogen is evolved when they react with water.

**B** Ions of Group I elements have a –1 charge.

**C** Sodium is more reactive than potassium.

**D** Solid sodium is a poor electrical conductor.

**23** Osmium is a transition element.

Which row gives the expected properties of osmium?

| | melting point | density | compounds formed |
|---|---|---|---|
| **A** | high | high | coloured |
| **B** | high | high | white |
| **C** | high | low | white |
| **D** | low | high | coloured |

**24** Two statements about noble gases are given.

1 Noble gases are reactive, monatomic gases.

2 Noble gases all have full outer shells of electrons.

Which is correct?

**A** Both statements are correct and statement 2 explains statement 1.

**B** Both statements are correct but statement 2 does not explain statement 1.

**C** Statement 1 is correct but statement 2 is incorrect.

**D** Statement 2 is correct but statement 1 is incorrect.

ROOT 19


www.root19.com +91 - 9969 353 391

**25** Some properties of substance X are listed.

- It conducts electricity when molten.
- It has a high melting point.
- It burns in oxygen and the product dissolves in water to give a solution with pH 11.

What is X?

**A** a covalent compound

**B** a macromolecule

**C** a metal

**D** an ionic compound

**26** Four metals P, Q, R and S are added to separate aqueous solutions of their ions.

The results are shown.

| metal | $P^{2+}$ | $Q^{2+}$ | $R^{2+}$ | $S^{2+}$ |
|---|---|---|---|---|
| P | ✗ | ✗ | ✓ | ✓ |
| Q | ✓ | ✗ | ✓ | ✓ |
| R | ✗ | ✗ | ✗ | ✗ |
| S | ✗ | ✗ | ✓ | ✗ |

key
✓ = reaction occurs
✗ = reaction does not occur

What is the order of reactivity of the metals, most reactive first?

**A** Q → P → S → R

**B** Q → S → P → R

**C** R → P → S → Q

**D** R → S → P → Q

**27** Copper is a transition element used to make saucepans.

Which property is **not** correct for copper?

**A** good conductor of heat

**B** insoluble in water

**C** low melting point

**D** malleable (can be hammered into shape)


www.root19.com +91 - 9969 353 591

**28** Aluminium is extracted by electrolysis of a mixture of aluminium oxide and cryolite.

Which statement is **not** correct?

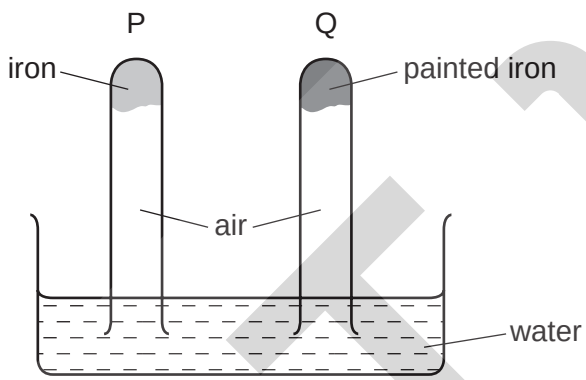
**A** The electrodes are made from graphite.

**B** The formula for aluminium oxide is $Al_2O_3$.

**C** The purpose of the cryolite is to lower the melting point of the mixture.

**D** The reaction taking place at the anode is $Al^{3+} + 3e^- \rightarrow Al$.

**29** The diagram shows an experiment to investigate how paint affects the rusting of iron.

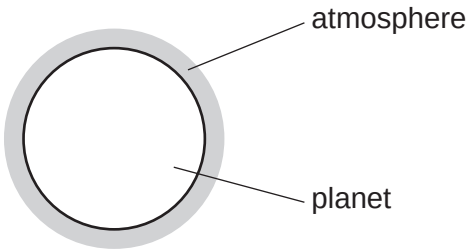



What happens to the water level in tubes P and Q?

| | tube P | tube Q |
|---|---|---|
| **A** | falls | rises |
| **B** | no change | rises |
| **C** | rises | falls |
| **D** | rises | no change |


www.root19.com +91 - 9969 353 391

**30** A new planet has been discovered and its atmosphere has been analysed.




The table shows the composition of its atmosphere.

| gas | percentage by volume |
|---|---|
| carbon dioxide | 4 |
| nitrogen | 72 |
| oxygen | 24 |

Which gases are present in the atmosphere of the planet in a higher percentage than they are in the Earth's atmosphere?

**A** carbon dioxide and oxygen

**B** carbon dioxide only

**C** nitrogen and oxygen

**D** nitrogen only

**31** Catalytic converters are used to remove some gaseous pollutants from car exhaust fumes.

Which gas is removed from the fumes by oxidation?

**A** carbon dioxide

**B** carbon monoxide

**C** nitrogen

**D** nitrogen oxide

**32** Ammonia is produced by the Haber process.

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g).$$

Which statement about the Haber process is **not** correct?

**A** An iron catalyst is used to increase the rate of reaction.

**B** The reaction is carried out at high temperature to increase the rate of reaction.

**C** The reaction is carried out at low pressure to increase the yield of ammonia.

**D** The reaction is reversible.

**33** One step in the manufacture of sulfuric acid is the oxidation of sulfur dioxide to sulfur trioxide.

Which conditions are used for this step?

| | temperature / °C | pressure / atmospheres | catalyst |
|---|---|---|---|
| **A** | 450 | 1.5 | iron |
| **B** | 450 | 1.5 | vanadium(V) oxide |
| **C** | 450 | 200 | iron |
| **D** | 450 | 200 | vanadium(V) oxide |

**34** Which process is used to make lime (calcium oxide) from limestone (calcium carbonate)?

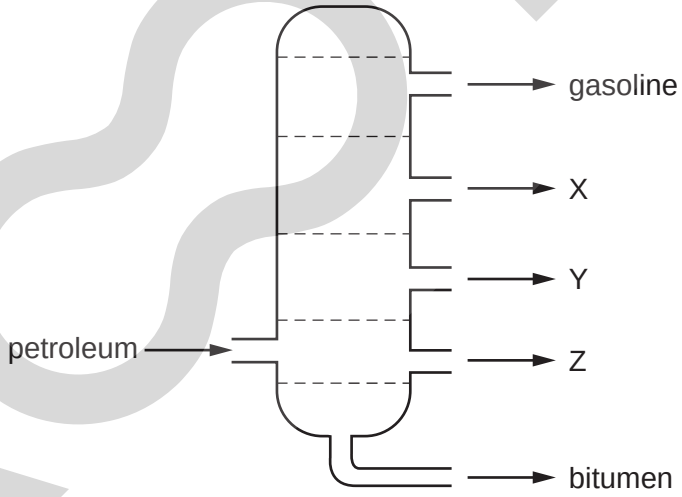
**A** chromatography

**B** electrolysis

**C** fractional distillation

**D** thermal decomposition

**35** The diagram shows the separation of petroleum into fractions.




What could X, Y and Z represent?

| | X | Y | Z |
|---|---|---|---|
| **A** | diesel oil | lubricating fraction | paraffin |
| **B** | lubricating fraction | diesel oil | paraffin |
| **C** | paraffin | lubricating fraction | diesel oil |
| **D** | paraffin | diesel oil | lubricating fraction |

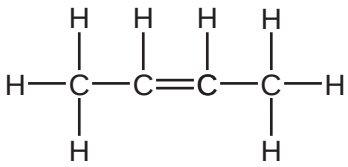

www.root19.com +91 - 9969 353 391

**36** Which compound does **not** belong to the same homologous series as the other three compounds?

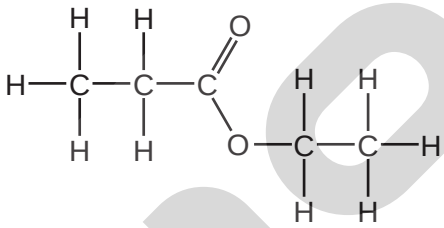
**A** $CH_3OH$ **B** $C_2H_5COOH$ **C** $C_2H_5OH$ **D** $C_7H_{15}OH$

**37** The structure of an alkene and the structure of an ester are shown.

P



Q



What are the names of P and Q?

| | P | Q |
|---|---|---|
| **A** | but-1-ene | ethyl propanoate |
| **B** | but-1-ene | propyl ethanoate |
| **C** | but-2-ene | ethyl propanoate |
| **D** | but-2-ene | propyl ethanoate |

**38** What is an advantage of producing ethanol by fermentation of sugar compared to the catalytic addition of steam to ethene?

**A** The alcohol produced is purer.

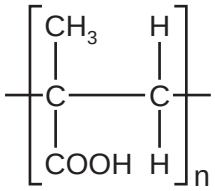
**B** The process is faster.

**C** The process uses high temperature.

**D** The process uses renewable raw materials.

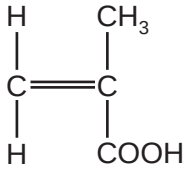
root 19

**39** A polymer has the formula shown.

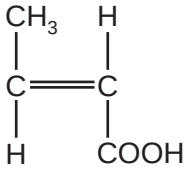


From which monomers can it be formed?

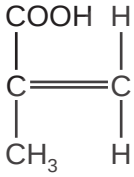
R



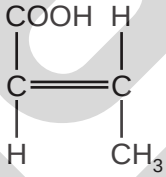
S



T



U



**A** R and S **B** R and T **C** S and U **D** T and U

**40** Which row shows a natural polymer with the same linkages as a synthetic polymer?

| | natural polymer | synthetic polymer |
|---|---|---|
| **A** | complex carbohydrate | nylon |
| **B** | complex carbohydrate | *Terylene* |
| **C** | protein | nylon |
| **D** | protein | *Terylene* |


www.root19.com +91 - 9969 353 391

**BLANK PAGE**

© UCLES 2016
www.root19.com

+91 - 9969 353 391

**BLANK PAGE**

© UCLES 2016
www.root19.com

+91 - 9969 353 391

**BLANK PAGE**

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.



www.root19.com +91 - 9969 353 391

## The Periodic Table of Elements

**Key**

atomic number
atomic symbol
name
relative atomic mass

| Group I | Group II | | | | | | | | | | | Group III | Group IV | Group V | Group VI | Group VII | Group VIII |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | 1<br>H<br>hydrogen<br>1 | | | | | | | | | | 2<br>He<br>helium<br>4 |
| 3<br>Li<br>lithium<br>7 | 4<br>Be<br>beryllium<br>9 | | | | | | | | | | | 5<br>B<br>boron<br>11 | 6<br>C<br>carbon<br>12 | 7<br>N<br>nitrogen<br>14 | 8<br>O<br>oxygen<br>16 | 9<br>F<br>fluorine<br>19 | 10<br>Ne<br>neon<br>20 |
| 11<br>Na<br>sodium<br>23 | 12<br>Mg<br>magnesium<br>24 | | | | | | | | | | | 13<br>Al<br>aluminium<br>27 | 14<br>Si<br>silicon<br>28 | 15<br>P<br>phosphorus<br>31 | 16<br>S<br>sulfur<br>32 | 17<br>Cl<br>chlorine<br>35.5 | 18<br>Ar<br>argon<br>40 |
| 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40 | 21<br>Sc<br>scandium<br>45 | 22<br>Ti<br>titanium<br>48 | 23<br>V<br>vanadium<br>51 | 24<br>Cr<br>chromium<br>52 | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56 | 27<br>Co<br>cobalt<br>59 | 28<br>Ni<br>nickel<br>59 | 29<br>Cu<br>copper<br>64 | 30<br>Zn<br>zinc<br>65 | 31<br>Ga<br>gallium<br>70 | 32<br>Ge<br>germanium<br>73 | 33<br>As<br>arsenic<br>75 | 34<br>Se<br>selenium<br>79 | 35<br>Br<br>bromine<br>80 | 36<br>Kr<br>krypton<br>84 |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89 | 40<br>Zr<br>zirconium<br>91 | 41<br>Nb<br>niobium<br>93 | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>– | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103 | 46<br>Pd<br>palladium<br>106 | 47<br>Ag<br>silver<br>108 | 48<br>Cd<br>cadmium<br>112 | 49<br>In<br>indium<br>115 | 50<br>Sn<br>tin<br>119 | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128 | 53<br>I<br>iodine<br>127 | 54<br>Xe<br>xenon<br>131 |
| 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137 | 57–71<br>lanthanoids | 72<br>Hf<br>hafnium<br>178 | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184 | 75<br>Re<br>rhenium<br>186 | 76<br>Os<br>osmium<br>190 | 77<br>Ir<br>iridium<br>192 | 78<br>Pt<br>platinum<br>195 | 79<br>Au<br>gold<br>197 | 80<br>Hg<br>mercury<br>201 | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207 | 83<br>Bi<br>bismuth<br>209 | 84<br>Po<br>polonium<br>– | 85<br>At<br>astatine<br>– | 86<br>Rn<br>radon<br>– |
| 87<br>Fr<br>francium<br>– | 88<br>Ra<br>radium<br>– | 89–103<br>actinoids | 104<br>Rf<br>rutherfordium<br>– | 105<br>Db<br>dubnium<br>– | 106<br>Sg<br>seaborgium<br>– | 107<br>Bh<br>bohrium<br>– | 108<br>Hs<br>hassium<br>– | 109<br>Mt<br>meitnerium<br>– | 110<br>Ds<br>darmstadtium<br>– | 111<br>Rg<br>roentgenium<br>– | 112<br>Cn<br>copernicium<br>– | | 114<br>Fl<br>flerovium<br>– | | 116<br>Lv<br>livermorium<br>– | | |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| lanthanoids | 57<br>La<br>lanthanum<br>139 | 58<br>Ce<br>cerium<br>140 | 59<br>Pr<br>praseodymium<br>141 | 60<br>Nd<br>neodymium<br>144 | 61<br>Pm<br>promethium<br>– | 62<br>Sm<br>samarium<br>150 | 63<br>Eu<br>europium<br>152 | 64<br>Gd<br>gadolinium<br>157 | 65<br>Tb<br>terbium<br>159 | 66<br>Dy<br>dysprosium<br>163 | 67<br>Ho<br>holmium<br>165 | 68<br>Er<br>erbium<br>167 | 69<br>Tm<br>thulium<br>169 | 70<br>Yb<br>ytterbium<br>173 | 71<br>Lu<br>lutetium<br>175 |
| actinoids | 89<br>Ac<br>actinium<br>– | 90<br>Th<br>thorium<br>232 | 91<br>Pa<br>protactinium<br>231 | 92<br>U<br>uranium<br>238 | 93<br>Np<br>neptunium<br>– | 94<br>Pu<br>plutonium<br>– | 95<br>Am<br>americium<br>– | 96<br>Cm<br>curium<br>– | 97<br>Bk<br>berkelium<br>– | 98<br>Cf<br>californium<br>– | 99<br>Es<br>einsteinium<br>– | 100<br>Fm<br>fermium<br>– | 101<br>Md<br>mendelevium<br>– | 102<br>No<br>nobelium<br>– | 103<br>Lr<br>lawrencium<br>– |

The volume of one mole of any gas is 24 $dm^3$ at room temperature and pressure (r.t.p.)

© UCLES 2016

www.root19.com
+91 - 9969 353 391
PMT